MAPS FROM PSEUDO-COHERENT OBJECTS INTO SUPPORTED COMPLEXES KILL A PERFECT FUNCTION SUPPORTED OFF T

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-perfect.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof is complete: the reduction to O_X[0], the finite-stage realization and the cone construction

Editorial note. The label lemma-map-from-pseudo-coherent-to-complex-with-support names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- the local subcategory of D_QCoh supported on a closed subset and its perfect generator
- internal Hom in D_QCoh, its quasi-coherence and compatibility with restriction to opens
- colimit representations using the generator and compactness of O_X
- the factor-through lemma for maps into the thick subcategory generated by E
- perfectness of the cone over a map to a perfect object supported on T

Source reference. Original lemma 094A: sources/stacks-chow/tags.txt, source lines 11543–11543. Prerequisite pointer invoked at proof line(s) 4258; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 09SP: sources/stacks-chow/tags.txt, source lines 12362–12362. Prerequisite pointer invoked at proof line(s) 4259; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 09SN: sources/stacks-chow/tags.txt, source lines 12361–12361. Prerequisite pointer invoked at proof line(s) 4252; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0A9A: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 3997–3997. Prerequisite pointer invoked at proof line(s) 4248; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:spaces-perfect.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original reference 0A6H: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 3499–3499. Prerequisite pointer invoked at proof line(s) 4238; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let S be a scheme. Let X be a quasi-compact and quasi-separated algebraic space over S. Let $T \subset |X|$ be a closed subset such that the complement $U \subset X$ is quasi-compact. Let $\alpha : P \to E$ be a morphism of $D_{QCoh}(\mathcal{O}_X)$ with either*

(1) *P is perfect and E supported on T, or*
(2) *P pseudo-coherent, E supported on T, and E bounded below.*

Then there exists a perfect complex of $\mathcal{O}_X$-modules I and a map $I \to \mathcal{O}_X[0]$ such that $I \otimes^{\mathbf{L}} P \to E$ is zero and such that $I|_U \to \mathcal{O}_U[0]$ is an isomorphism.

Proof. Set $\mathcal{D} = D_{QCoh,T}(\mathcal{O}_X)$. In both cases the complex $K = R\mathcal{H}om(P, E)$ is an object of $\mathcal{D}$. See Lemma 0A6H for quasi-coherence. It is clear that K is supported on T as formation of $R\mathcal{H}om$ commutes with restriction to opens. The map α defines an element of $H^0(K) = \mathrm{Hom}_{D(\mathcal{O}_X)}(\mathcal{O}_X[0], K)$. Then it suffices to prove the result for the map $\alpha : \mathcal{O}_X[0] \to K$.

Let $E \in \mathcal{D}$ be a perfect generator, see Lemma 0A9A. Write

$$K = \mathrm{hocolim} K_n$$

as in Derived Categories, Lemma 09SN using the generator E. Since the functor $\mathcal{D} \to D(\mathcal{O}_X)$ commutes with direct sums, we see that $K = \mathrm{hocolim} K_n$ holds in $D(\mathcal{O}_X)$. Since $\mathcal{O}_X$ is a compact object of $D(\mathcal{O}_X)$ we find an n and a morphism $\alpha_n : \mathcal{O}_X \to K_n$ which gives rise to α, see Derived Categories, Lemma 094A. By Derived Categories, Lemma 09SP applied to the morphism $\mathcal{O}_X[0] \to K_n$ in the ambient category $D(\mathcal{O}_X)$ we see that α_n factors as $\mathcal{O}_X[0] \to Q \to K_n$ where Q is an object of $\langle E \rangle$. We conclude that Q is a perfect complex supported on T.

Choose a distinguished triangle

$$I \to \mathcal{O}_X[0] \to Q \to I[1]$$

By construction I is perfect, the map $I \to \mathcal{O}_X[0]$ restricts to an isomorphism over U, and the composition $I \to K$ is zero as α factors through Q. This proves the lemma. □